Implementation of Deques in Python , Java and C++

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Python

Collection.deque is built in deque class of python , it is a doubly linked list of fixed-size blocks. Not a linked list of items , a linked list of arrays. Each block holds 64 elements . Blocks have prev and next pointers linking them into the previous and next pointers .It provides fast insertion and deletion from both the head end and the tail end and can function as a queue and a stack. Application areas include scheduling, sliding algorithms, and real time processing systems.Python deques efficiently pop and append from both ends saving memory ,cache . Its designed to include double linked lists of fixed size blocks .

Java

java.util.ArrayDeque is designed as a linear collection of data . Default initial capacity is 16 however , the array doesnt have restrictions and grows according to need . Most ArrayDeque operations run in amortized constant time the exceptions to these operations are : remove(Object), contains(Object), iterator.remove(), and the bulk operations. One contiguous array means excellent cache behavior and O(1) indexing . The official Java documentation claims that it is better than stack as a stack and linkedlist as a queue .

C++

std::deque is the built in double-ended queue of C++, implemented as a dynamic array of pointer to fixed size heap-allocated blocks . not a linked list of items, and

not one contiguous array. Blocks have no pointers pointing to the previous and next so , their order comes from their position in the map, and on growth only the map is reallocated, so existing elements are never moved. It combines vector and queues because of its chunked layout , within each fixed-size block, elements are contiguous like a vector, giving O(1) random access and good cache behavior; across blocks, the map of pointers lets it grow at either end like a queue, so push/pop at front and back are O(1) and existing elements are never relocated.

References :

https://dev.to/3bdelrahman/exploring-1-based-indexing-in-programming-languages-169k
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https://www.geeksforgeeks.org/java/deque-interface-java-example/